CHM-DSM-401 — Fundamental of Chemistry-IV

Unit II: Renewable Energy

7th Semester, Assam University (FYUG) — Haflong Government College

Paper: CHM-DSM-401, Full Marks 100 [ESE 70 + Internal 30] | Pass marks 40 | Unit II: 9 lectures

Notes 17 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

Index

Unit syllabus (as in the syllabus file)

1. Solar energy: Sun as a source of energy, Solar radiation, Importance, Storage of solar energy, Different types of Solar cells.
2. Bio-Energy: Energy from biomass, Sources of biomass, Different species, Conversion of biomass into fuels, Energy through fermentation, Pyrolysis and gasification.
3. Hydrogen Energy: Production and Storage, Applications.

Top 2 high-mark topics in this unit

1. Different types of solar cells and storage of solar energy — syllabus names types explicitly, a descriptive diagram/process long answer, the classic long question in a minor paper.
2. Conversion of biomass into fuels - energy through fermentation, pyrolysis and gasification — three named processes in one syllabus line, ideal for a compare-and-contrast long answer.

Contents

1. Solar Energy: The Sun as a Source of Energy
2. Solar Radiation and Its Importance
3. Storage of Solar Energy and Types of Solar Cells
4. Bio-Energy: Biomass and Its Conversion into Fuels
5. Hydrogen Energy: Production, Storage and Applications
6. Questions — Short form, then Long form (no verbatim AU questions found for this unit)
7. Sources

1. Solar Energy: The Sun as a Source of Energy

Solar energy originates from thermonuclear fusion reactions taking place within the core of the Sun. Under conditions of extreme temperature ($\approx 1.5 \times 10^7$ K) and high pressure, hydrogen nuclei (protons) undergo nuclear fusion to form helium-4 nuclei (^4_2He).

The overall nuclear fusion reaction can be written as:

$$4\,^{1}_{1}\text{H} \longrightarrow \,^{4}_{2}\text{He} + 2\,^{0}_{1}e^{+} + 2\,\nu_e + \gamma$$

where ^1_1H represents a proton, ^4_2He is a helium nucleus, $^0_1e^+$ is a positron, ν_e is an electron neutrino, and γ represents high-energy gamma-ray photons.

The energy liberated during this fusion process arises from the mass defect (Δm), which is the difference between the total mass of the reacting protons and the mass of the resulting helium nucleus.

The mass of four individual protons is given by:

$$m_{4p} = 4 \times 1.007276 \text{ u} = 4.029104 \text{ u}$$

The mass of a single helium nucleus is:

$$m_{\text{He}} = 4.001506 \text{ u}$$

The mass defect Δm is defined as the difference between these two masses:

$$\Delta m = m_{4p} - m_{\text{He}}$$

Substituting the mass values into the mass defect expression yields:

$$\Delta m = 4.029104 \text{ u} - 4.001506 \text{ u}$$

Subtracting the values gives:

$$\Delta m = 0.027598 \text{ u}$$

Using the conversion factor $1 \text{ u} = 1.66054 \times 10^{-27}$ kg, the mass defect in kilograms is calculated as:

$$\Delta m = 0.027598 \times 1.66054 \times 10^{-27} \text{ kg}$$

Multiplying these terms gives:

$$\Delta m = 4.5828 \times 10^{-29} \text{ kg}$$

According to Einstein's mass-energy equivalence relationship, the energy released (ΔE) per fusion event is:

$$\Delta E = \Delta m c^2$$

Substituting the value of the mass defect $\Delta m = 4.5828 \times 10^{-29}\ \text{kg}$ and the speed of light $c = 2.998 \times 10^{8}\ \text{m s}^{-1}$ yields:

$$\Delta E = (4.5828 \times 10^{-29}\ \text{kg}) \times (2.998 \times 10^{8}\ \text{m s}^{-1})^2$$

Squaring the speed of light:

$$\Delta E = (4.5828 \times 10^{-29}\ \text{kg}) \times (8.988 \times 10^{16}\ \text{m}^2\ \text{s}^{-2})$$

Evaluating the product yields the energy in Joules:

$$\Delta E = 4.119 \times 10^{-12}\ \text{J}$$

Converting Joules to Megaelectronvolts via $1\ \text{eV} = 1.602 \times 10^{-19}\ \text{J}$:

$$\Delta E = \frac{4.119 \times 10^{-12}\ \text{J}}{1.602 \times 10^{-13}\ \text{J MeV}^{-1}} \approx 25.7\ \text{MeV}$$

The two positrons produced in the fusion reaction annihilate with two electrons, releasing a further $2 \times 1.022 = 2.04$ MeV, so the total energy of the reaction is about 26.7 MeV (about 26.1 MeV of it stays as heat and light; the rest is carried away by the neutrinos).

Continuous fusion inside the solar core results in total radiated power (solar luminosity, P_{sun}) of approximately:

$$P_{\text{sun}} \approx 3.8 \times 10^{26}\ \text{W}$$

The Solar Constant

The solar energy radiated into space travels radially outward in all directions. The intensity of this radiation at the average distance between the Earth and the Sun, outside the atmosphere, is defined as the **Solar Constant** (G_{sc}).

The solar constant is defined as the rate at which solar radiant energy is received per unit area on a surface perpendicular to the direction of solar rays at the mean Earth-Sun distance outside the Earth's atmosphere.

Let R_{ES} represent the mean radius of Earth's orbit around the Sun:

$$R_{ES} \approx 1.496 \times 10^{11}\ \text{m}$$

Assuming spherical symmetry, the solar power P_{sun} is distributed over a sphere of surface area A:

$$A = 4\pi R_{ES}^2$$

Substituting the orbit radius into the surface area formula:

$$A = 4 \times \pi \times (1.496 \times 10^{11}\ \text{m})^2$$

Squaring the orbital distance:

$$A = 4 \times \pi \times (2.238 \times 10^{22}\ \text{m}^2)$$

Multiplying numerical factors gives:

$$A = 2.812 \times 10^{23}\ \text{m}^2$$

The solar constant G_{sc} is defined as:

$$G_{sc} = \frac{P_{\text{sun}}}{A}$$

Substituting $P_{\text{sun}} = 3.844 \times 10^{26}\ \text{W}$ and $A = 2.812 \times 10^{23}\ \text{m}^2$:

$$G_{sc} = \frac{3.844 \times 10^{26}\ \text{W}}{2.812 \times 10^{23}\ \text{m}^2}$$

Performing the division yields the standard value:

$$G_{sc} \approx 1367\ \text{W m}^{-2}$$

The Solar Spectrum and Atmospheric Interactions

The electromagnetic spectrum emitted by the Sun approximates that of a blackbody operating at a surface temperature of approximately $5778\ \text{K}$. Outside the Earth's atmosphere, the extraterrestrial solar spectrum is divided into three primary wavelength bands:

1. **Ultraviolet (UV) Radiation ($\lambda < 400\ \text{nm}$):** Constitutes approximately 8% of the total solar energy.
2. **Visible Light ($400\ \text{nm} \leq \lambda \leq 700\ \text{nm}$):** Accounts for approximately 43% of total solar energy, peaking near $500\ \text{nm}$.
3. **Infrared (IR) Radiation ($\lambda > 700\ \text{nm}$):** Accounts for approximately 49% of total solar energy.

As solar radiation penetrates the Earth's atmosphere, its spectral composition and overall intensity are modified through three primary physical processes: absorption, scattering, and reflection.

1. Atmospheric Absorption

- Stratospheric ozone (O_3) absorbs almost all short-wavelength ultraviolet radiation ($\lambda < 290\ \text{nm}$) and a large portion of UV-B radiation ($290\ \text{nm} - 320\ \text{nm}$). - Atmospheric water vapor (H_2O) and carbon dioxide (CO_2) absorb selective wavelength bands in the infrared region.

2. Atmospheric Scattering

- **Rayleigh Scattering:** Occurs when solar radiation interacts with gas molecules smaller than the radiation wavelength. The scattering cross-section is inversely proportional to the fourth power of wavelength ($\propto \lambda^{-4}$), scattering shorter blue wavelengths more efficiently. - **Mie Scattering:** Occurs when radiation interacts with atmospheric aerosols, dust particles, and water droplets whose sizes are comparable to or larger than the wavelength of light.

3. Atmospheric Reflection and Albedo

- Clouds reflect a significant fraction of incoming solar radiation back into space. - **Albedo:** The fraction of total incident solar radiation reflected by the Earth-atmosphere system. The Earth's average planetary albedo is approximately 0.30 (or 30%).

Because of these atmospheric interactions, approximately $50\% - 70\%$ of the extraterrestrial solar radiation reaches the Earth's surface, depending on atmospheric path length and cloud coverage.

2. Solar Radiation and Its Importance

Direct and Diffuse Radiation

Solar radiation reaching the Earth's surface consists of two distinct components:

1. **Direct (Beam) Radiation (G_{direct}):** Solar radiation that travels in a straight line from the Sun to the Earth's surface without undergoing scattering by atmospheric gases, aerosols, or clouds. Direct radiation casts sharp shadows.
2. **Diffuse Radiation (G_{diffuse}):** Solar radiation whose direction has been randomized by scattering processes in the atmosphere before arriving at the surface. Diffuse radiation originates from all directions in the sky vault and does not cast sharp shadows.

The total solar radiation incident on a horizontal surface at the Earth's surface is termed **Global Radiation** (G_{global}). It is expressed mathematically as:

$$G_{\text{global}} = G_{\text{direct}} \cos\theta_z + G_{\text{diffuse}}$$

where θ_z is the solar zenith angle (the angle between the Sun's rays and the vertical normal to the horizontal surface).

Insolation and Factors Affecting Its Intensity

Insolation (Incoming Solar Radiation) is defined as the total solar radiation energy received per unit surface area over a specified time interval (typically expressed in W m^{-2} or $\text{kWh m}^{-2}\ \text{day}^{-1}$).

The magnitude of insolation received at any point on the Earth's surface depends on several parameters:

1. Latitude

The angle of incidence of solar rays varies with geographic latitude. At the equator, solar rays strike the surface nearly perpendicularly, distributing energy over a smaller surface area and passing through a shorter atmospheric path length. At higher latitudes, solar rays strike at an oblique angle, spreading energy over a larger area and passing through a thicker layer of atmosphere:

$$I_{\text{surface}} = I_0 \cos\theta$$

where I_0 is the incident beam intensity and θ is the angle of incidence.

2. Season

The Earth's rotational axis is tilted at an angle of approximately 23.5° relative to its orbital plane around the Sun. This axial tilt alters the solar declination angle throughout the year, changing both the solar elevation angle at noon and the day length (photoperiod) for any given location.

3. Time of Day

The rotation of the Earth causes the solar zenith angle (θ_z) to vary continuously from sunrise to sunset. Insolation is zero at night, increases during the morning, reaches its maximum at local solar noon when θ_z is minimized, and decreases through the afternoon.

4. Cloud Cover and Atmospheric Conditions

Clouds reflect and absorb incoming solar radiation, significantly decreasing the direct radiation component while increasing the proportion of diffuse radiation. High relative humidity and atmospheric pollutant concentrations also attenuate total insolation.

Average Solar Energy Received

The global average solar radiation reaching the Earth's surface on a horizontal plane ranges between $160\ \mathrm{W\ m^{-2}}$ and $250\ \mathrm{W\ m^{-2}}$ when averaged over 24 hours. Expressed as daily energy yield, tropical and arid regions receive between $5\ \mathrm{kWh\ m^{-2}\ day^{-1}}$ and $7\ \mathrm{kWh\ m^{-2}\ day^{-1}}$, while temperate regions receive between $3\ \mathrm{kWh\ m^{-2}\ day^{-1}}$ and $4.5\ \mathrm{kWh\ m^{-2}\ day^{-1}}$.

Importance of Solar Energy

Solar energy possesses key characteristic advantages as a primary clean energy resource:

- **Abundant and Inexhaustible:** The solar energy striking the Earth's surface in one hour exceeds the total global energy consumption of human civilization in an entire year. - **Renewable and Free Resource:** Solar radiation is naturally replenished continuously and carries zero fuel acquisition cost. - **Environmentally Clean:** Solar energy conversion systems emit zero direct carbon dioxide ($\mathrm{CO_2}$), sulfur dioxide ($\mathrm{SO_2}$), nitrogen oxides (NO_x), or particulate matter during operation, significantly lowering greenhouse gas emissions. - **Ubiquitous Availability:** Solar radiation is available to varying degrees across all geographic regions globally, reducing geopolitical energy dependencies. - **Decentralized and Off-Grid Suitability:** Solar technologies can operate at modular scales, making them suited for powering remote, off-grid, and rural communities where grid infrastructure expansion is economically unviable. - **Fossil Fuel Independence:** Utilization of solar energy conserves finite fossil fuel reserves (coal, petroleum, and natural gas) for non-combustion chemical feedstocks.

Limitations of Solar Energy

Despite its abundance, solar energy utilization presents several intrinsic technical challenges:

- **Intermittent Availability:** Solar radiation varies periodically due to the diurnal (day-night) cycle and seasonally due to Earth's orbital geometry. Solar energy is completely unavailable during night hours. - **Diffuse Nature:** Solar radiation exhibits a low power density ($\leq 1000\ \mathrm{W\ m^{-2}}$ peak at ground level), requiring expansive surface collection areas (collectors or photovoltaic arrays) to capture substantial quantities of energy. - **Dependence on Weather Conditions:** Unpredictable atmospheric conditions, such as cloud cover, rain, snow,

and atmospheric dust, cause fluctuations in energy output. - **Requirement for Storage Infrastructure:** To ensure a reliable, continuous power supply despite intermittency, solar systems require secondary energy storage systems (such as chemical batteries or thermal storage media), which increases capital costs.

3. Storage of Solar Energy and Types of Solar Cells

Need for Solar Energy Storage

Solar radiation arriving at the Earth's surface is inherently intermittent and variable. The diurnal cycle introduces a zero-input period during the night, while meteorological fluctuations, such as cloud cover and seasonal changes, cause significant variation in solar flux density throughout the day. Furthermore, peak power generation from solar collection occurs around solar noon, whereas energy consumption peaks typically occur during early morning and evening hours.

To bridge this operational discrepancy between energy supply and demand, effective energy storage systems are essential. Storage systems stabilize grid operations, maintain power continuity, and allow captured solar energy to be dispatched on demand.

Methods of Solar Energy Storage

Solar energy can be converted and stored through thermal, electrical, and chemical pathways.

Thermal Energy Storage

Thermal storage systems capture thermal energy directly from solar collectors or solar concentrating systems for domestic, industrial, or power generation applications.

Sensible Heat Storage

Sensible heat storage involves raising the temperature of a solid or liquid medium without undergoing a phase change. The total heat energy Q stored in a material of mass m and specific heat capacity C_p over a temperature differential from T_1 to T_2 is expressed by the thermal equation

$$Q = \int_{T_1}^{T_2} mC_p \, dT$$

Assuming the specific heat capacity C_p remains constant over the working temperature range $\Delta T = T_2 - T_1$, this integral simplifies to

$$Q = mC_p\Delta T$$

Water is the standard liquid storage medium for low-to-medium temperature applications owing to its exceptionally high specific heat capacity ($C_p \approx 4.184 \text{ J g}^{-1}\text{K}^{-1}$). For high-temperature sensible storage (such as in concentrated solar power plants), packed rock beds, concrete blocks, or molten nitrate salts are utilized due to their high thermal stability at elevated temperatures.

Latent Heat Storage

Latent heat storage utilizes Phase-Change Materials (PCMs) that store or release energy during a thermodynamic phase transition at a constant transformation temperature. The total energy Q stored during solid-to-liquid melting of a mass m of PCM involves both sensible heating of the solid phase and the latent enthalpy of fusion ΔH_{fusion}, given by

$$Q = mC_{p,\text{solid}}(T_{\text{melt}} - T_1) + m\Delta H_{\text{fusion}} + mC_{p,\text{liquid}}(T_2 - T_{\text{melt}})$$

Phase-change materials offer high energy density over a narrow temperature interval. Common PCMs include organic compounds (such as paraffin waxes and fatty acids) and inorganic compounds (such as salt hydrates like hydrated sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$).

Solar Ponds

A solar pond is a body of saltwater designed to collect and store solar thermal energy by suppressing natural thermal convection. Standard water bodies develop natural convection currents: as bottom layers heat up, their density decreases, causing them to rise and lose heat to the atmosphere.

A salinity-gradient solar pond prevents convection by dissolving salt (such as sodium chloride or magnesium chloride) near the bottom, creating a vertical density gradient. The pond is structured into three distinct horizontal zones:

1. **Upper Convective Zone (UCZ):** A thin top layer of fresh or low-salinity water at ambient temperature.
2. **Non-Convective Zone (NCZ):** A middle gradient layer where salt concentration and temperature both increase with depth. This layer acts as a transparent thermal insulator.
3. **Lower Convective Zone (LCZ):** A high-salinity layer at the bottom that acts as the thermal collector and heat storage zone, capable of reaching temperatures of 80°C to 90°C.

Electrical Energy Storage

Solar electrical storage systems temporarily store direct electrical output from photovoltaic panels into electrochemical or mechanical devices.

Lead-Acid Batteries

Lead-acid storage cells consist of a lead dioxide (PbO_2) cathode and a spongy lead (Pb) anode immersed in an aqueous sulfuric acid (H_2SO_4) electrolyte. During discharge, both electrodes are converted into lead sulfate (PbSO_4).

The half-reaction occurring at the spongy lead anode during discharge is

$$\text{Pb}(s) + \text{HSO}_4^-(aq) \rightarrow \text{PbSO}_4(s) + \text{H}^+(aq) + 2e^-$$

The half-reaction occurring at the lead dioxide cathode during discharge is

$$\text{PbO}_2(s) + 3\text{H}^+(aq) + \text{HSO}_4^-(aq) + 2e^- \rightarrow \text{PbSO}_4(s) + 2\text{H}_2\text{O}(l)$$

Combining the anode and cathode half-reactions yields the overall reversible cell discharge reaction

$$\mathrm{Pb}(s) + \mathrm{PbO_2}(s) + 2\mathrm{H_2SO_4}(aq) \rightleftharpoons 2\mathrm{PbSO_4}(s) + 2\mathrm{H_2O}(l)$$

Reversing the current direction via solar input drives the reaction backward, regenerating Pb and $\mathrm{PbO_2}$.

Lithium-Ion Batteries

Lithium-ion batteries operate via host-guest intercalation of lithium ions ($\mathrm{Li^+}$) between host matrices. A typical cell comprises a graphitic carbon anode ($\mathrm{C_6}$) and a lithium transition metal oxide cathode (such as $\mathrm{LiCoO_2}$).

During charging via photovoltaic power, lithium ions deintercalate from the cathode host lattice and intercalate into the graphitic carbon structure. The cathode half-reaction is

$$\mathrm{LiCoO_2} \rightarrow \mathrm{Li_{1-x}CoO_2} + x\mathrm{Li^+} + xe^-$$

The corresponding anode half-reaction is

$$x\mathrm{Li^+} + xe^- + \mathrm{C_6} \rightarrow \mathrm{Li}_x\mathrm{C_6}$$

The total reversible cell process during charge and discharge is represented by

$$\mathrm{LiCoO_2} + \mathrm{C_6} \rightleftharpoons \mathrm{Li_{1-x}CoO_2} + \mathrm{Li}_x\mathrm{C_6}$$

Lithium-ion batteries offer superior energy density, higher round-trip efficiency ($\approx 90\%$), and longer cycle life compared to lead-acid batteries.

Pumped Hydroelectric Storage

Pumped hydro systems store electrical energy mechanically as gravitational potential energy. Excess photovoltaic power generated during peak sunshine hours drives electric pumps to transport water from a lower reservoir to an elevated upper reservoir.

The theoretical gravitational potential energy E_p stored by lifting a mass of water m through an effective vertical height elevation h under gravitational acceleration g is

$$E_p = mgh$$

During periods of high electricity demand or low solar illumination, stored water flows downward through hydraulic turbines connected to electrical generators, returning energy to the grid with a round-trip efficiency of approximately $70 - 80\%$.

Chemical Energy Storage

Chemical energy storage involves using solar energy to break low-energy chemical bonds and synthesize high-energy molecular fuels ("solar fuels").

Water Electrolysis for Hydrogen Production

Direct solar-to-chemical conversion uses solar-derived electrical current (or direct photo-electrochemical energy) to split water into molecular hydrogen and oxygen. The overall endothermic electrochemical decomposition of liquid water is

$$2\mathrm{H_2O}(l) \xrightarrow{\text{Electrical / Solar Energy}} 2\mathrm{H_2}(g) + \mathrm{O_2}(g)$$

In an acidic proton-exchange membrane (PEM) electrolyzer, the oxidation half-reaction occurring at the anode is the Oxygen Evolution Reaction (OER)

$$2\mathrm{H_2O}(l) \rightarrow \mathrm{O_2}(g) + 4\mathrm{H^+}(aq) + 4e^-$$

The reduction half-reaction occurring at the cathode is the Hydrogen Evolution Reaction (HER)

$$4\mathrm{H^+}(aq) + 4e^- \rightarrow 2\mathrm{H_2}(g)$$

Molecular hydrogen has a high gravimetric energy density ($\approx 120\ \mathrm{MJ\ kg^{-1}}$) and can be stored in compressed gas tanks, cryo-liquid systems, or metal hydride matrices for later power generation with zero greenhouse gas emissions.

Solar Fuels

Solar fuels encompass chemical species produced by catalytic reduction processes driven by solar radiation. This includes reducing carbon dioxide with water to produce synthetic hydrocarbons, methanol ($\mathrm{CH_3OH}$), or syngas ($\mathrm{CO + H_2}$) via artificial photosynthesis and photo-catalytic systems.

Photovoltaic Effect and Solar Cells

The Photovoltaic Effect

The photovoltaic effect is the physical and chemical process by which a semiconductor material directly converts light energy into electricity. When a photon with energy equal to or greater than the semiconductor's bandgap energy (E_g) strikes the material, it is absorbed, promoting an electron from the valence band to the conduction band and creating an electron-hole pair.

The photon energy E is related to frequency ν and wavelength λ by

$$E = h\nu = \frac{hc}{\lambda}$$

where h is Planck's constant and c is the speed of light. The condition for electron-hole generation is

$$h\nu \geq E_g$$

If a p-n junction is present within the semiconductor, a built-in electric field is established across the depletion region due to diffusion of mobile charge carriers (electrons from the n-side diffuse to the p-side, and holes from the p-side diffuse to the n-side).

When light-generated electron-hole pairs reach this internal electric field region, the field sweeps free electrons toward the n-type layer and free holes toward the p-type layer. This charge separation prevents recombination and establishes a photovoltage across the junction terminals. Connecting an external load completes the circuit, driving electric current.

The power conversion efficiency η of a solar cell measures its capacity to convert incident solar irradiance into usable electrical power, given by

$$\eta = \frac{P_{\text{max}}}{P_{\text{in}}} = \frac{V_{\text{oc}}\, I_{\text{sc}}\, \text{FF}}{P_{\text{in}}}$$

where P_{max} is the maximum output electrical power, P_{in} is the total incident solar power, V_{oc} is the open-circuit voltage, I_{sc} is the short-circuit current, and FF is the fill factor of the solar cell's current-voltage (I-V) output curve.

Types of Solar Cells

Monocrystalline Silicon Solar Cells

Monocrystalline silicon cells are fabricated from single-crystal silicon ingots, usually grown via the Czochralski process. The crystal lattice of the resulting wafer is continuous throughout the entire material, free of grain boundaries.

Because grain boundaries are absent, charge-carrier recombination losses are low, resulting in high electron mobility and commercial efficiencies typically between 15% and 20%. However, the energy-intensive Czochralski growth process and the precise slicing required make monocrystalline silicon relatively expensive.

Polycrystalline Silicon Solar Cells

Polycrystalline (or multicrystalline) silicon cells are produced by melting raw silicon and pouring it into square molds, where it slowly cools and crystallizes. This structure consists of multiple distinct silicon grains separated by microscopic crystal boundaries.

The presence of grain boundaries creates structural defects that trap charge carriers and increase recombination rates. Consequently, polycrystalline cells exhibit slightly lower power conversion efficiencies ($\approx 13\% - 16\%$). However, the simpler manufacturing process makes them less expensive to produce than monocrystalline cells.

Thin-Film Solar Cells

Thin-film solar cells are fabricated by depositing micro-meter thick layers of photovoltaic material onto lightweight rigid or flexible substrates (such as glass, stainless steel foil, or polymers). Their reduced material thickness significantly decreases production weight and cost.

Amorphous Silicon (a-Si)

Amorphous silicon lacks long-range crystalline order. Because non-crystalline silicon has a higher light absorption coefficient than crystalline silicon, thin films ($< 1\ \mu\text{m}$) are sufficient to absorb visible solar light. However, structural disorder causes higher defect states and lower efficiencies ($\approx 6\% - 10\%$), alongside light-induced degradation (the Staebler-Wronski effect).

Cadmium Telluride (CdTe)

Cadmium telluride is a direct bandgap compound semiconductor ($E_g \approx 1.45\ \text{eV}$) tailored to the solar emission spectrum. It offers high absorption efficiency with thin layers, achieving commercial efficiencies of $10\% - 15\%$. Production costs are low, though toxicity concerns surrounding cadmium require strict end-of-life recycling.

Copper Indium Gallium Selenide (CIGS)

CIGS is a quaternary direct bandgap semiconductor ($\text{CuIn}_{1-x}\text{Ga}_x\text{Se}_2$) with an adjustable bandgap ($1.0\ \text{eV}$ to $1.7\ \text{eV}$, depending on the Ga/In ratio). CIGS cells achieve the highest efficiency among thin-film technologies ($\approx 12\% - 16\%$) and show minimal degradation over time, but manufacturing requires precise stoichometric control during deposition.

Dye-Sensitized Solar Cells (DSSCs)

Dye-sensitized solar cells (also known as Grätzel cells) are photo-electrochemical systems that separate the functions of light absorption and charge transport. A standard DSSC consists of:

1. A photo-anode consisting of a mesoporous, high-surface-area metal oxide semiconductor film (typically titanium dioxide, TiO_2) deposited on a transparent conducting oxide glass substrate.
2. Synthetic organic or organometallic dye molecules (such as ruthenium polypyridyl complexes) adsorbed onto the surface of the TiO_2 nanoparticles.
3. A redox electrolyte containing an iodide/triiodide (I^-/I_3^-) couple dissolved in an organic solvent.
4. A platinum-coated counter electrode.

When light strikes the cell, the adsorbed dye molecule (D) absorbs a photon and enters an excited state (D^*):

$$D + h\nu \rightarrow D^*$$

The excited dye molecule injects an electron into the conduction band of the adjacent TiO_2 matrix:

$$D^* \rightarrow D^+ + e^-(\text{CB, TiO}_2)$$

The injected electron diffuses through the nanostructured TiO_2 network to the transparent conducting front electrode, moving through the external electrical load to the counter electrode.

Meanwhile, the oxidized dye molecule (D^+) is regenerated to its neutral ground state by accepting an electron from iodide ions in the electrolyte:

$$2D^+ + 3\text{I}^- \rightarrow 2D + \text{I}_3^-$$

The triiodide ions (I_3^-) generated in this step diffuse across to the counter electrode, where they are reduced back to iodide by electrons arriving from the external circuit:

$$\text{I}_3^- + 2e^- \rightarrow 3\text{I}^-$$

DSSCs perform well under low light intensity, ambient indoor light, and diffuse sunlight, though long-term electrolyte stability remains a key development area.

Perovskite Solar Cells

Perovskite solar cells utilize absorber materials featuring a perovskite crystal lattice structure with the general formula ABX_3, where A is a monovalent organic or inorganic cation (such as methylammonium $\mathrm{CH_3NH_3^+}$, formamidinium $\mathrm{HC(NH_2)_2^+}$, or $\mathrm{Cs^+}$), B is a divalent metal cation (typically $\mathrm{Pb^{2+}}$ or $\mathrm{Sn^{2+}}$), and X is a halide anion ($\mathrm{I^-}$, $\mathrm{Br^-}$, or $\mathrm{Cl^-}$). A common example is methylammonium lead triiodide ($\mathrm{CH_3NH_3PbI_3}$).

Perovskite absorbers feature high absorption coefficients, long charge-carrier diffusion lengths, tunable bandgaps, and high defect tolerance. They can be processed from solution at low temperatures, making them low-cost to produce. However, challenges remain regarding their operational stability when exposed to atmospheric moisture, ultraviolet light, and heat.

Multi-Junction (Tandem) Solar Cells

Multi-junction solar cells stack multiple single-junction solar cells made of different semiconductor materials with progressively smaller bandgaps on top of one another.

In a tandem arrangement, the top cell has the widest bandgap ($E_{g,1}$) and absorbs high-energy, short-wavelength photons while allowing lower-energy photons to pass through. The underlying middle and bottom cells have smaller bandgaps ($E_{g,2}$ and $E_{g,3}$) that absorb progressively longer wavelengths of light.

For example, a triple-junction cell might stack indium gallium phosphide (InGaP, $E_g \approx 1.8\ \mathrm{eV}$), gallium arsenide ($\mathrm{GaAs}$, $E_g \approx 1.4\ \mathrm{eV}$), and germanium ($\mathrm{Ge}$, $E_g \approx 0.67\ \mathrm{eV}$).

By capturing different regions of the solar emission spectrum, multi-junction cells minimize thermalization losses (where photon energy exceeding E_g is wasted as heat). These devices achieve the highest conversion efficiencies available ($> 30\%$ under unconcentrated sunlight, and $> 45\%$ under concentrated sunlight). Due to high fabrication costs, multi-junction cells are primarily used in space applications (such as satellite power systems) and concentrated photovoltaic (CPV) arrays.

Comparison of Solar Cell Technologies

Solar Cell Type	Active Material(s)	Typical Efficiency ($\%$)	Relative Cost	Primary Applications
Monocrystalline Silicon	Single-crystal silicon (Si)	$15\% - 20\%$	High	Residential rooftops, commercial solar farms, space panels
Polycrystalline Silicon	Multi-grain silicon (Si)	$13\% - 16\%$	Moderate	Utility-scale ground arrays, rooftop installations
Thin-Film (a-Si)	Amorphous silicon ($\mathrm{a\text{-}Si{:}H}$)	$6\% - 10\%$	Low	Small consumer electronics (calculators, watches), building integration

Solar Cell Type	Active Material(s)	Typical Efficiency (%)	Relative Cost	Primary Applications
Thin-Film (CdTe)	Cadmium Telluride (CdTe)	$10\% - 15\%$	Low	Utility-scale solar power plants
Thin-Film (CIGS)	Copper Indium Gallium Selenide	$12\% - 16\%$	Moderate	Flexible modules, Building-Integrated Photovoltaics (BIPV)
Dye-Sensitized (DSSC)	Ruthenium dye / $\mathrm{TiO_2}$ / $\mathrm{I^-/I_3^-}$	$7\% - 11\%$	Low	Indoor sensors, low-power electronics, smart windows
Perovskite	Organometallic halides (e.g., $\mathrm{CH_3NH_3PbI_3}$)	$18\% - 25\%$	Low (potential)	Tandem configurations, next-generation commercial modules
Multi-Junction (Tandem)	Stacked III-V semiconductors (e.g., $\mathrm{InGaP/GaAs/Ge}$)	$30\% - > 45\%$	Very High	Aerospace, satellites, concentrated photovoltaics (CPV)

4. Bio-Energy: Biomass and Its Conversion into Fuels

4.1 Energy from Biomass

Bio-energy refers to the chemical energy stored in organic matter, known as biomass, derived from recently living organisms such as plants, animals, and microorganisms. Biomass represents a form of stored solar energy. Through the fundamental biological process of photosynthesis, green plants capture electromagnetic radiation from the sun and convert inorganic carbon dioxide and water into energy-rich organic compounds, predominantly carbohydrates. The chemical equation representing the conversion of solar radiant energy into stored chemical energy in glucose is written as:

$$6\mathrm{CO_2} + 6\mathrm{H_2O} \xrightarrow{h\nu} \mathrm{C_6H_{12}O_6} + 6\mathrm{O_2}$$

The energy stored within the chemical bonds of synthesized carbohydrates ($\mathrm{C_6H_{12}O_6}$) can subsequently be released through direct thermochemical processing or biochemical conversion pathways to yield heat, electricity, or gaseous and liquid fuels.

4.2 Sources of Biomass

Biomass feedstocks can be derived from several agricultural, forestry, municipal, and aquatic streams:

- **Agricultural Residues:** These are non-food plant materials left behind after crop harvesting and processing. Prominent examples include rice straw, sugarcane bagasse (the fibrous residue remaining after sugar extraction), and corn stover (the leaves, stalks, and cobs of maize). - **Forestry Residues and Wood:** Waste

materials produced during forest management, logging operations, and timber processing, including tree branches, treetops, sawdust, bark, and mill offcuts. - **Energy Crops:** Dedicated crops cultivated specifically for bio-energy production on a large scale. These include herbaceous energy crops like switchgrass (*Panicum virgatum*) and miscanthus (*Miscanthus* spp.), as well as sugar-rich crops like sugarcane and sweet sorghum. - **Animal Waste and Dung:** Excreta from livestock (cattle, swine, and poultry) that contain high proportions of biodegradable organic matter suitable for biological conversion. - **Municipal Solid Waste (MSW):** The organic fraction of household and commercial waste, including food scraps, yard waste, and paper residues. - **Algae:** Aquatic organisms, including both microscopic microalgae and macroscopic macroalgae (seaweeds), capable of rapid growth and high biomass productivity per unit area.

4.3 Different Species Used as Energy Crops

Specific plant and algal species are selected for bio-energy production based on their growth rates, yield of target chemical components (such as sugars, starches, lipids, or lignocellulose), and suitability for specific energy conversion processes:

- **Sugarcane (*Saccharum officinarum*):** A C4 perennial grass cultivated extensively in tropical and subtropical regions. Its high content of fermentable sucrose makes it the primary crop for industrial bioethanol production. - **Jatropha (*Jatropha curcas*) and Pongamia (*Pongamia pinnata*):** Non-edible oilseed plants that produce seeds rich in lipids (triglycerides). The extracted seed oils serve as key non-food feedstocks for transesterification to produce biodiesel. - **Eucalyptus (*Eucalyptus* spp.) and Poplar (*Populus* spp.):** Fast-growing short-rotation woody crops cultivated for their high lignocellulosic biomass production. They are primarily harvested as solid wood fuel for direct combustion, gasification, or pelletization. - **Algae:** Microalgal species (such as *Chlorella* and *Scenedesmus*) store large amounts of lipids within their cellular structures, which are extracted for biodiesel production, while the residual algal biomass undergo anaerobic digestion to yield biogas.

4.4 Conversion of Biomass into Fuels

Biomass can be converted into solid, liquid, and gaseous fuels through thermochemical, biochemical, and chemical routes.

Direct Combustion

Direct combustion is the oxidation of biomass in the presence of excess atmospheric oxygen at high temperatures ($800^\circ\text{C} - 1000^\circ\text{C}$) to produce thermal energy. The liberated heat is used directly for space heating, industrial processes, or to raise steam in a boiler to drive a steam turbine for electricity generation. The ideal combustion reaction of biomass carbohydrates is expressed as:

$$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \Delta H$$

Energy Through Fermentation

Fermentation involves the biological degradation of organic substrates by microorganisms under specific biochemical environments.

Alcoholic Fermentation

Simple hexose sugars derived from sugarcane juice, sweet sorghum, or hydrolyzed starches are converted into bioethanol and carbon dioxide through the metabolic activity of yeast species, primarily *Saccharomyces cerevisiae*, under anaerobic conditions:

$$\mathrm{C_6H_{12}O_6} \xrightarrow{\text{yeast}} 2\mathrm{C_2H_5OH} + 2\mathrm{CO_2}$$

The resulting aqueous ethanol mixture is purified by fractional distillation and dehydration to obtain fuel-grade anhydrous ethanol ($\mathrm{C_2H_5OH}$).

Anaerobic Digestion

Wet biomass, such as animal dung, sewage sludge, and organic municipal waste, undergoes biochemical breakdown by a consortium of anaerobic bacteria (acidogens, acetogens, and methanogenic archaea) in an oxygen-free environment. The overall process degrades complex organic polymers into biogas, a gaseous mixture consisting primarily of methane ($\mathrm{CH_4}$, 50–70%) and carbon dioxide ($\mathrm{CO_2}$, 30–50%), along with trace amounts of hydrogen sulfide ($\mathrm{H_2S}$). A representative balanced reaction for methane generation from glucose is given by:

$$\mathrm{C_6H_{12}O_6} \rightarrow 3\mathrm{CH_4} + 3\mathrm{CO_2}$$

Pyrolysis

Pyrolysis is the thermal decomposition of biomass conducted in the total absence of oxygen at elevated temperatures. Depending on the heating rate, residence time, and temperature range, pyrolysis converts solid biomass into liquid bio-oil, solid biochar, and non-condensable syngas.

- **Slow Pyrolysis:** Conducted at low temperatures ($300^\circ\mathrm{C} - 500^\circ\mathrm{C}$) with slow heating rates and long vapor residence times. The primary product is solid, carbon-rich **biochar**. - **Fast Pyrolysis:** Conducted at intermediate temperatures ($450^\circ\mathrm{C} - 600^\circ\mathrm{C}$) with rapid heating rates ($> 1000^\circ\mathrm{C/s}$) and short vapor residence times (< 2 seconds). The process maximizes the yield of liquid **bio-oil** (up to 75 wt%), an oxygenated liquid fuel mixture. - **Gas-Phase / Flash Pyrolysis:** Conducted at high temperatures ($> 700^\circ\mathrm{C}$), maximizing the yield of non-condensable **syngas**.

Gasification

Gasification is the thermochemical conversion of biomass into a combustible gaseous mixture through partial oxidation at high temperatures ($700^\circ\mathrm{C} - 1000^\circ\mathrm{C}$) using a sub-stoichiometric amount of an oxidizing agent such as air, pure oxygen, or steam.

The initial step involves partial oxidation of biomass carbon:

$$\mathrm{C} + \frac{1}{2}\mathrm{O_2} \rightarrow \mathrm{CO}$$

This reaction is followed by the endothermic steam-reforming reaction of carbon:

$$\mathrm{C} + \mathrm{H_2O} \rightarrow \mathrm{CO} + \mathrm{H_2}$$

Additionally, the water-gas shift reaction establishes an equilibrium between the gas species:

$$\mathrm{CO} + \mathrm{H_2O} \rightleftharpoons \mathrm{CO_2} + \mathrm{H_2}$$

The resulting gaseous mixture, known as synthesis gas or syngas, consists mainly of carbon monoxide (CO) and hydrogen ($\mathrm{H_2}$). Syngas can be burned directly in gas turbines for electricity generation or utilized as a chemical feedstock to produce synthetic liquid hydrocarbon fuels via the Fischer-Tropsch catalytic reaction:

$$(2n+1)\mathrm{H_2} + n\mathrm{CO} \xrightarrow{\text{catalyst}} \mathrm{C}_n\mathrm{H}_{2n+2} + n\mathrm{H_2O}$$

Transesterification of Vegetable Oils

Biodiesel consists of fatty acid methyl esters (FAME) produced by the transesterification of triglycerides derived from non-edible seed oils (such as *Jatropha* or *Pongamia*) or algal lipids. In this chemical transformation, a triglyceride molecule reacts with a short-chain alcohol, typically methanol ($\mathrm{CH_3OH}$), in the presence of a strong base catalyst (such as sodium hydroxide, NaOH, or potassium hydroxide, KOH) or an acid catalyst.

The chemical reaction requires three moles of methanol per mole of triglyceride to yield three moles of fatty acid methyl ester (biodiesel) and one mole of glycerol as a byproduct. Representing the triglyceride with long-chain fatty acid alkyl groups $\mathrm{R_1}$, $\mathrm{R_2}$, and $\mathrm{R_3}$:

$$\begin{array}{l} \mathrm{CH_2-O-CO-R_1} \\ \quad | \\ \mathrm{CH-O-CO-R_2} \\ \quad | \\ \mathrm{CH_2-O-CO-R_3} \end{array} + 3\mathrm{CH_3OH} \xrightarrow{\text{catalyst}} \begin{array}{l} \mathrm{R_1-COOCH_3} \\ \quad + \\ \mathrm{R_2-COOCH_3} \\ \quad + \\ \mathrm{R_3-COOCH_3} \end{array} + \begin{array}{l} \mathrm{CH_2-OH} \\ \quad | \\ \mathrm{CH-OH} \\ \quad | \\ \mathrm{CH_2-OH} \end{array}$$

The resulting biodiesel ($\mathrm{R\text{-}COOCH_3}$) possesses viscosity and combustion characteristics comparable to petroleum-derived diesel, allowing its direct use in standard compression-ignition engines.

5. Hydrogen Energy: Production, Storage and Applications

Hydrogen is an efficient, clean energy carrier with high gravimetric energy density. Because hydrogen does not occur in large quantities as a free gas in nature, it must be produced from hydrogen-containing compounds such as water, natural gas, or biomass using primary energy sources.

Production of Hydrogen

Steam Methane Reforming of Natural Gas

Steam Methane Reforming (SMR) is currently the predominant industrial method for bulk hydrogen production. Methane reacts endothermically with high-temperature steam ($700^\circ\mathrm{C}$ to $1000^\circ\mathrm{C}$) under pressures of 3 to 25 bar in the presence of a nickel catalyst to form carbon monoxide and hydrogen:

$$\mathrm{CH_4(g)} + \mathrm{H_2O(g)} \rightarrow \mathrm{CO(g)} + 3\mathrm{H_2(g)}$$

To maximize hydrogen yield, the gaseous mixture undergoes the exothermic water-gas shift (WGS) reaction, in which carbon monoxide reacts with additional steam over iron oxide or copper catalysts:

$$\mathrm{CO(g) + H_2O(g) \rightarrow CO_2(g) + H_2(g)}$$

Combining the reforming and shift steps yields the overall stoichiometric equation:

$$\mathrm{CH_4(g) + 2H_2O(g) \rightarrow CO_2(g) + 4H_2(g)}$$

Electrolysis of Water

Electrolysis involves splitting water molecules into elemental hydrogen and oxygen by passing a direct electric current through an aqueous electrolyte or proton-exchange membrane. The overall endothermic decomposition reaction is given by:

$$\mathrm{2H_2O(l) \rightarrow 2H_2(g) + O_2(g)}$$

At the cathode, reduction of water or hydrogen ions generates hydrogen gas:

$$\mathrm{2H^+(aq) + 2e^- \rightarrow H_2(g)}$$

At the anode, oxidation of water yields oxygen gas and protons:

$$\mathrm{2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-}$$

When the electricity supplied to the electrolyzer is derived entirely from renewable sources (such as solar or wind power), the process yields zero direct carbon emissions.

Biomass Gasification

Biomass gasification converts organic matter into a synthesis gas (syngas) comprising carbon monoxide, carbon dioxide, hydrogen, and methane under high temperatures ($700^\circ\mathrm{C}$ to $1000^\circ\mathrm{C}$) with a controlled amount of oxygen or steam. Representing biomass as a general carbohydrate unit $\mathrm{C}_n\mathrm{H}_m\mathrm{O}_k$, the partial oxidation and steam reforming reactions generate syngas:

$$\mathrm{C}_n\mathrm{H}_m\mathrm{O}_k + \frac{n-k}{2}\mathrm{O_2} \rightarrow n\mathrm{CO} + \frac{m}{2}\mathrm{H_2}$$

$$\mathrm{C}_n\mathrm{H}_m\mathrm{O}_k + (n-k)\mathrm{H_2O} \rightarrow n\mathrm{CO} + \left(n + \frac{m}{2} - k\right)\mathrm{H_2}$$

Subsequent clean-up and water-gas shift processing of the produced carbon monoxide increases the yield of hydrogen gas.

Photoelectrochemical Water Splitting

Photoelectrochemical (PEC) water splitting directly utilizes solar radiation to decompose water into hydrogen and oxygen without intermediate electricity generation. A semiconductor photoelectrode absorbs incident photons with energy greater than its bandgap, generating electron-hole pairs. The photogenerated holes oxidize water at the photoanode surface:

$$2H_2O(l) + 4h^+ \rightarrow O_2(g) + 4H^+(aq)$$

The photogenerated electrons migrate through an external circuit to the cathode to reduce hydrogen ions to gaseous hydrogen:

$$4H^+(aq) + 4e^- \rightarrow 2H_2(g)$$

Colour Labels of Hydrogen

Hydrogen is categorized by colour codes based on its production pathways and environmental impact: * **Grey Hydrogen:** Produced from fossil fuels (primarily steam methane reforming or coal gasification) without capturing the resulting carbon dioxide emissions. * **Blue Hydrogen:** Produced via fossil fuel processes (such as steam methane reforming) where the carbon dioxide emissions are captured and stored using Carbon Capture and Storage (CCS) technologies. * **Green Hydrogen:** Produced through water electrolysis using 100% renewable electricity, or through direct photoelectrochemical splitting, resulting in near-zero life-cycle carbon emissions.

Storage of Hydrogen

High-Pressure Compressed Gas Cylinders

Hydrogen gas has a very low volumetric density at ambient conditions. To store practical quantities, it is compressed to pressures ranging from 350 bar (used in heavy duty transport) to 700 bar (used in light duty passenger vehicles). These systems rely on high-strength composite materials (such as carbon-fiber reinforced polymers with polymer liners) to withstand structural fatigue and prevent gas leakage.

Cryogenic Liquid Hydrogen Storage

Liquefying hydrogen increases its volumetric density significantly compared to gaseous storage. Hydrogen undergoes liquefaction at ambient pressure by cooling it down to $-253^\circ C$ ($20\ K$). Cryogenic tanks require double-walled vacuum-insulated vessels to minimize heat ingress and boil-off losses. Liquid storage is advantageous for bulk transport, long-distance logistics, and aerospace applications.

Metal Hydrides

Hydrogen can react reversibly with certain metals and metallic alloys to form solid metal hydrides. Intermetallic alloys such as lanthanum-nickel ($LaNi_5$) store hydrogen atoms within their interstitial lattice sites at moderate pressures and room temperature:

$$LaNi_5(s) + 3H_2(g) \rightleftharpoons LaNi_5H_6(s)$$

The absorption process is exothermic, releasing heat during charging, while the desorption process is endothermic, requiring heat input to release pure hydrogen gas. This method offers high volumetric energy density and safe operation at low pressure.

Chemical Carriers

Hydrogen can be chemically bonded into liquid or easily liquefiable molecules for safe, ambient-condition transport: * **Ammonia (NH_3):** Produced by reacting hydrogen with atmospheric nitrogen via the Haber-Bosch

process. Ammonia liquefies at moderate pressure (around 8.5 bar at 20°C) or low cooling (-33°C), providing high volumetric hydrogen capacity. It can be cracked back into H_2 and N_2 at the point of use. * **Liquid Organic Hydrogen Carriers (LOHCs):** Unsaturated organic compounds (such as toluene or dibenzyltoluene) undergo catalytic hydrogenation to form stable, liquid saturated molecules (such as methylcyclohexane). The hydrogenated carrier can be safely stored and transported using existing liquid fuel infrastructure before being catalytically dehydrogenated at its destination:

$$\text{C}_7\text{H}_8(\text{l}) + 3\text{H}_2(\text{g}) \rightleftharpoons \text{C}_7\text{H}_{14}(\text{l})$$

Comparison of Storage Methods

Storage Method	Volumetric Energy Density	Safety Considerations	Cost and Energy Efficiency
Compressed Gas (350–700 bar)	Moderate to low	High pressure poses mechanical stress and puncture risks	Compression requires approximately 10–15% of the fuel's energy content; moderate cylinder cost
Liquid Hydrogen (-253°C)	High	Extremely cold temperatures require cryogenic handling; boil-off pressure buildup risks	Liquefaction requires up to 30% of the lower heating value of hydrogen; high insulation container cost
Metal Hydrides (LaNi_5)	High volumetric capacity	Highly stable; operates at low pressure, low risk of explosive release	High system weight; significant thermal management required; high alloy material cost
Chemical Carriers (Ammonia/LOHC)	Very high volumetric capacity	Ammonia is toxic; LOHCs are generally non-toxic and non-explosive under ambient conditions	Requires energy input for catalytic cracking or dehydrogenation; lower overall round-trip efficiency

Applications of Hydrogen Energy

Fuel Cells

Fuel cells convert the chemical energy of hydrogen directly into electrical energy through electrochemical reactions, bypassing the thermodynamic Carnot cycle efficiency limitations of thermal combustion engines. In a Proton Exchange Membrane Fuel Cell (PEMFC), hydrogen oxidation takes place at the anode:

$$\text{H}_2(\text{g}) \rightarrow 2\text{H}^+(\text{aq}) + 2\text{e}^-$$

Protons travel through the selective polymer membrane to the cathode, while electrons travel through an external electrical circuit to generate direct current. At the cathode, oxygen reduction occurs:

$$\frac{1}{2}\mathrm{O_2(g)} + 2\mathrm{H^+(aq)} + 2\mathrm{e^-} \rightarrow \mathrm{H_2O(l)}$$

Combining the two half-reactions gives the overall cell reaction:

$$\mathrm{H_2(g)} + \frac{1}{2}\mathrm{O_2(g)} \rightarrow \mathrm{H_2O(l)}$$

The only direct operational byproduct of this process is pure water and heat.

Transport

Hydrogen is utilized in fuel cell electric vehicles (FCEVs) including cars, buses, heavy-duty trucks, trains, and marine vessels. Hydrogen transportation provides long driving ranges, rapid refueling times comparable to conventional liquid fossil fuels, and zero tailpipe greenhouse gas emissions.

Industry

Hydrogen serves as a critical chemical feedstock and reducing agent across several major industrial sectors: * **Ammonia Synthesis:** Hydrogen is combined with nitrogen in the Haber-Bosch process to produce ammonia ($\mathrm{NH_3}$), the fundamental precursor for nitrogen fertilizers. * **Petroleum Refining:** Hydrotreating removes impurities such as sulfur from crude oil fractions, while hydrocracking breaks heavy hydrocarbons into lighter, high-value fuels. * **Steel Making:** Green hydrogen replaces carbon-intensive metallurgical coke as a direct reducing agent for iron ore ($\mathrm{Fe_2O_3}$), reducing the oxide to direct reduced iron (DRI):

$$\mathrm{Fe_2O_3(s)} + 3\mathrm{H_2(g)} \rightarrow 2\mathrm{Fe(s)} + 3\mathrm{H_2O(g)}$$

This process emits water vapor instead of carbon dioxide.

Energy Storage and Grid Balancing

Intermittent renewable energy production from solar and wind power can be converted into green hydrogen via water electrolysis during periods of excess generation. The stored hydrogen can later be re-converted to electricity using fuel cells or gas turbines during periods of peak demand, balancing the electrical grid across daily and seasonal timescales.

Heating and Cooking

Hydrogen can be blended into existing natural gas distribution networks or utilized directly in dedicated pure hydrogen burners for residential space heating, commercial hot water supply, and high-temperature industrial thermal processes, offering an alternative to direct fossil fuel combustion.

Questions

Short form first (2–3 marks), then long form (5–10 marks). **No clean verbatim Assam University question was found for this unit** — only two paraphrased topic matches from CHMSEC-401T Fuel Chemistry (jumbled extraction, exact wording cannot be verified), marked [AU] below. The other questions are syllabus-derived, marked as such.

Short-form questions

S1. [AU, paraphrased] Write a brief note on solar energy. — CHMSEC-401T Fuel Chemistry, 4th Sem, 2024, CBCS (Unit II). Note: jumbled multi-column extraction; exact wording cannot be verified.

Answer in brief: Sections 1–2 — the Sun radiates about 3.8×10^{26} W, solar constant about 1367 W/m^2; solar energy is abundant, renewable and emission-free.

S2. [AU, paraphrased] Write a brief note on biodiesel. — CHMSEC-401T Fuel Chemistry, 4th Sem, 2024, CBCS (Unit III). Note: jumbled multi-column extraction; exact wording cannot be verified.

Answer in brief: Section 4 — biodiesel is made by transesterification of vegetable oils (for example jatropha or pongamia seed oil) with methanol; it is a renewable diesel substitute.

S3. [standard] What is the solar constant? Name the main types of solar cells.

Answer in brief: Sections 1 and 3 — about 1367 W/m^2 at the top of the atmosphere; monocrystalline and polycrystalline silicon, thin-film (CdTe, CIGS), dye-sensitized, perovskite and multi-junction cells.

S4. [standard] Name the products of pyrolysis and of gasification of biomass.

Answer in brief: Section 4 — pyrolysis gives bio-oil, biochar and syngas; gasification gives syngas (carbon monoxide and hydrogen).

Long-form questions

L1. [standard] Describe the Sun as a source of energy, solar radiation, importance of solar energy and storage of solar energy.

What to write: Sections 1–3 in full — fusion in the Sun, the solar constant and spectrum, direct and diffuse radiation, why solar energy matters, and thermal, battery and chemical storage.

L2. [standard] Describe different types of solar cells, with description, and compare them.

What to write: Section 3 in full — the photovoltaic effect, silicon, thin-film, dye-sensitized, perovskite and multi-junction cells, and the comparison table.

L3. [standard] What are the sources of biomass and different species used? Describe conversion of biomass into fuels.

What to write: Section 4 in full — agricultural and forestry residues, energy crops (sugarcane, jatropha, algae), and combustion, fermentation, pyrolysis, gasification and transesterification.

L4. [standard] Describe production and storage of hydrogen energy and state its applications.

What to write: Section 5 in full — steam reforming, electrolysis, biomass gasification; compressed gas, liquid hydrogen, metal hydrides and chemical carriers; fuel cells, transport, industry and grid storage.

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.